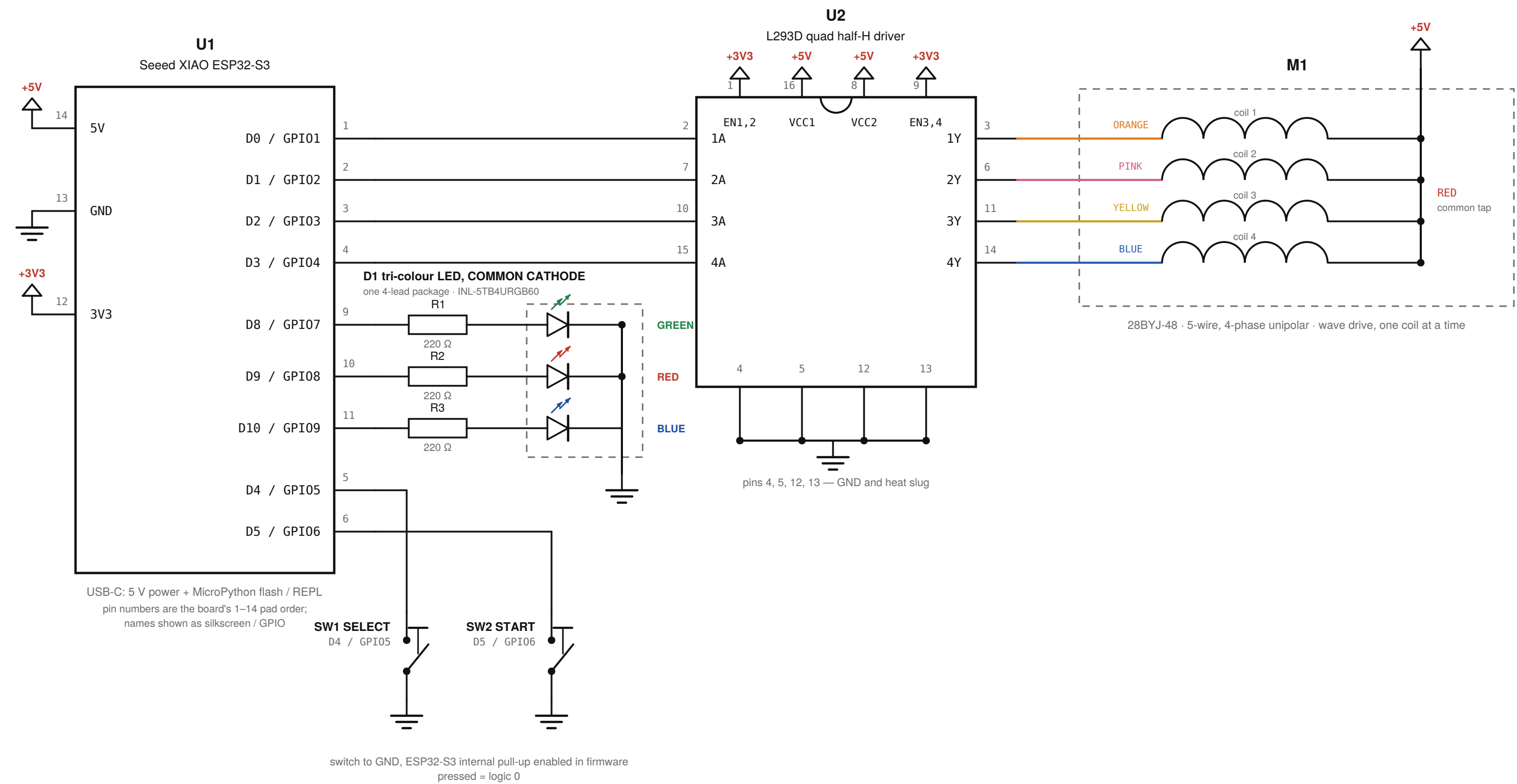


SELF-CONTAINED MEETING TIMER

Seeed XIAO ESP32-S3 · L293D quad half-H driver · 28BYJ-48 unipolar stepper · tri-colour LED · 2 × tactile switch



NOTES

- Pin names are given as silkscreen / GPIO. The XIAO's pads are printed D0–D10, not GPIO numbers — GPIO7/8/9 are the pads marked D8/D9/D10. Wire from the silkscreen name. Source: Seeed XIAO ESP32-S3 pinout.
- L293D VCC1 (pin 16) and VCC2 (pin 8) at +5 V; EN1,2 (pin 1) and EN3,4 (pin 9) at +3V3. Datasheet: VCC1 recommended 4.5–7.0 V, VIH(min) = 2.3 V, so 3.3 V is a valid logic high on the enables but would be out of spec on VCC1.
- GPIO logic is 3.3 V. No level shifter on the L293D inputs; the shift to the 5 V motor rail happens at the driver outputs.
- M1 is unipolar: the four half-windings share the red centre tap, which sits at +5 V. A coil therefore energises when its driver output is pulled LOW. firmware/config.py STEP_SEQUENCE is written active-HIGH — verify on the bench (see VERIFICATION.md).
- D1 is a single 4-lead common-cathode package, confirming COMMON_ANODE = False in firmware/leds.py. Green and blue VF = 2.8–3.6 V, so from 3.3 V through 220 Ω they draw only ~1–2 mA against red's ~6 mA; expect a large brightness imbalance.
- LED channel order is NOT red/blue/green by GPIO number: GPIO7 drives GREEN, GPIO8 drives RED, GPIO9 drives BLUE. This is the bench-verified assignment in firmware/config.py on lead/final (LED_GREEN = 7, LED_RED = 8, LED_BLUE = 9). Rev B and the wiring table in hardware/electrical/README.md still show the original guess and are stale.
- Not supplied in the kit, recommended: 100 nF ceramic across VCC1–GND at U2, and 100 μF bulk on the +5 V rail near pin 8.

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Drawn by: Deniz Oge	Rev C
Checked against firmware/config.py	2026-09-19